Indicative price for a 10-year solar PPA (Borrentin, 114 MWp)

Memo · 23 Oct 2026 · ppa-lab 1.12.0 · illustrative, from public data: not an offer, no view on actual contracts · details: Step 7 report

Bottom line. For a 10-year pay-as-produced PPA (2027-2036) the negotiation range is **21.4 to 35.1 EUR/MWh**: from the indicative price a buyer can accept after its risk premium and costs, up to the fair value before risk. At the indicative price the PPA cuts our 95% cash flow at risk from EUR 3.02 m to 0.70 m per year (**–77%**) for about **14.6 EUR/MWh** (EUR 1.77 m per year) against selling at spot.

The price, layer by layer (EUR/MWh)

Layer	Solar park	Wind park
Forward 2027 (average of the scenarios)	141.1	141.1
Shape: output falls in cheap hours	–63.5	–11.1
= Fair price 2027	**77.5**	**129.9**
Later years: lower forwards and capture rates; degradation	–42.4	–43.3
= Fair price, 10 years	**35.1**	**86.6**
Buyer's risk premium after a baseload hedge	–10.5	–5.5
Balancing and route to market (assumed)	–2.2	–1.6
Credit and liquidity (assumed)	–1.0	–1.0
= Indicative price, 10 years	**21.4**	**78.6**

What we keep and what we give away

- **Volume** stays with us: P50 120.9 GWh a year, one year in ten below 113.4 GWh (P90, 31 weather years).
- **Price and shape** go to the buyer. Its premium is mostly profile risk for solar (how deep the midday dip is) and volume risk for wind; the price level it can hedge with forwards.
- Against spot sales we give away the premium, the credit cost and the 3.1 EUR/MWh that stopping at negative prices saves (balancing is paid either way).

What would change the price

- Build-out scenarios for solar and storage after 2027: **–3.4 / +8.7**; ±1 pp a year of trend on top: **±3.5** EUR/MWh
- Forward curve –10% / +10%: **–5.6 / +5.5 EUR/MWh**
- Capture rate as far off as the model was out of sample in earlier years (+0.03, Step 6): **–2.1 EUR/MWh**

For comparison, the risk premium and the costs together take 13.7 EUR/MWh off the fair price.

Recommendations

1. Quote within the range above; do not go below 21.4 EUR/MWh.
2. Offer a negative-price clause as the alternative: we stop producing at negative prices (18% of the solar output in 2027, more later) and are paid 39.3 EUR/MWh per delivered MWh: 27.3 per MWh of output against 21.4 without the clause, at a higher cash flow at risk (EUR 1.29 m instead of 0.70 m).
3. Agree the cannibalisation view with Risk before signing, and for any unsold volume sell about 87% of the average output as Cal base futures: it cuts the merchant cash flow at risk by 72% (solar park).

Method and limits

496 joint scenarios for 2027: 31 ERA5 weather years at the real sites, each with simulated hourly prices that respond to that weather (calibrated on 2024-2026 German day-ahead prices, analog days from 2025-10-01). Fair price = E[Σ q P] / E[Σ q]; premium = 5% shortfall of the buyer's hedged value per MWh; later years by the forward curve, capture rates moving with the price level as in the model (and 2023-2026 history), solar's with the build-out, discounted at 6%. Forward: EEX German Power Cal-27 base 140.21 EUR/MWh (quote of 2026-10-09, via tacto.ai); its quarters follow the market's seasonal shape (Q1-Q4 185.3 / 117.7 / 124.3 / 134.3, quotes of 2026-09-14); later years follow the forward curve (2028: 104.75, 2029: 87.58), held flat after 2029. Limits: the forwards are published quotes from secondary sources, not EEX files, the costs are illustrative, grid curtailment is not modelled, and the shape comes from one market regime (2024-2026). The Cal-27 forward moved between 104 and 140 EUR/MWh since 2026-07-31 (published quotes): refresh the offer with the forward curve on the day of signing.